# HARMILAP SINGH DHALIWAL

*AI Engineering Lead · Agentic AI & Enterprise Systems*

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AI engineering lead with 8+ years building applied ML and production AI systems. Conceived, architected and built the bank's first true agentic AI platform approved for production, launched organization-wide in May 2026. Hands-on across orchestration, semantic retrieval, human review and evaluation, with technical direction, team development and cross-functional delivery.

## PROFESSIONAL EXPERIENCE

### AI & Data Science Lead, Finance | Royal Bank of Canada

*April 2025 - Present*

*CFO Group, Enterprise Finance · Toronto, ON*

*Remain ~70% hands-on across architecture, modeling, full-stack implementation, evaluation, and production operations while leading cross-functional delivery.*

#### Production AI Systems

- Built and launched the bank's first production-approved agentic AI platform, owning the core application stack across ETL, document ingestion, PostgreSQL/pgvector dense retrieval, a reviewable authoring workflow, and a single-agent LangGraph orchestrator with bounded MCP tool routines. Piloted in April 2026; launched organization-wide in May 2026.
- Conceived, architected, and built the AI/LLM Workforce Analytics Platform from March through its November 2025 production launch, leading cross-functional delivery. It supports authorized headcount, compensation-cost, and open-position questions across ~40,000 cost centres; model stages route intent while deterministic Python retrieves and calculates within the resolved entitlement scope.
- Led a focused two-week v1-to-v2 refactor of the production Financial Peer Benchmarking Platform: dense semantic retrieval ranks candidate metrics; LLM-assisted disambiguation resolves near-duplicate names; reviewed templates and parameter binding constrain text-to-SQL execution.

#### Technical Leadership

- Set architecture, code-review, and refactoring standards for retrieval, model behavior, and SQL guardrails; manage one Senior AI Scientist, guided the 2025 Amplify cohort, and helped junior contributors ship production changes.
- Own application evaluation and post-release follow-through: prepare LLM-as-judge evidence and extensive human-test results for Model Risk review, backed by custom tracing and logging.

### Enterprise Finance Senior Data Scientist | Royal Bank of Canada

*September 2022 - April 2025*

*CFO Group, Enterprise Finance · Toronto, ON*

- Built v1 of the Financial Peer Benchmarking Platform end to end, automating text extraction and entity matching across quarterly-changing disclosures from Canada's six largest banks; removed a long-standing peer-analysis bottleneck and earned the 2025 CFO One RBC Team Award.
- Cut commodity-tax return preparation from months to 90 minutes by building a PySpark pipeline supporting ~$600M in allocations from enterprise general-ledger data; earned the CFO Group RBC Quarterly Team Award (Q4 2023).

## EARLIER EXPERIENCE

### Machine Learning Engineer | Quantiphi Inc.

*October 2021 - September 2022*

*Toronto, ON*

- Improved checkbox-detection accuracy from ~70% to 99.95% on Humana claims documents by layering OpenCV pixel-level detection and Random Forest classification over OCR.
- Built document-classification and entity-extraction workflows and deployed custom and AutoML models on Vertex AI; developed supporting data infrastructure with Bigtable and BigQuery.

### Data Scientist | Tata Consultancy Services

*August 2016 - November 2019*

*Pune, India*

- Delivered $3M in annual savings for a 900 MW generating unit at Kansai Electric's 1,800 MW Maizuru coal-fired station in Japan. As sole data scientist on a three-person R&D team, built the combustion-tuning digital twin end-to-end: ETL across 90+ sensors, 84 regression models, and Particle Swarm Optimization that produced operator-reviewed boiler-tuning recommendations.
- Designed the model-selection framework using k-fold cross-validation, R-squared, RMSE, MAPE, and fold variance; also shipped LSTM forecasting and XGBoost transformer-life prediction models.
- Placed 2nd of 600 in TCS's computer-vision hackathon with a CNN math-notation detector; earned the Innovation Pride Award (2019).

## TECHNICAL SKILLS

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| **Programming & Production** | Python, SQL, PySpark, ETL, API service design, PostgreSQL/pgvector, Git/GitHub, observability; production delivery through partner-owned OpenShift CI/CD |
| **Machine Learning & Evaluation** | PyTorch, TensorFlow, scikit-learn, XGBoost; regression, classification, clustering, time series, CNNs, Transformers; model selection and evaluation |
| **Retrieval & LLM Systems** | Document classification and extraction, named entities, embeddings, dense retrieval, semantic ranking, near-duplicate disambiguation, dense RAG, guarded text-to-SQL, LangGraph, MCP, human-in-the-loop workflows, LLM evaluation |
| **Cloud & Data** | Google Cloud Platform, Vertex AI, AutoML, BigQuery, Bigtable |

## EDUCATION

**Post Graduate Professional Certificate, Big Data Analytics** — Georgian College · 2021

**B.Eng., Electronics & Communications Engineering** — Thapar University · 2016

## AWARDS & RECOGNITION

**CFO Conference Award** | RBC — 2026

**CFO Individual Quarterly Award** | RBC — P4 2026

**CFO One RBC Team Award** | RBC — 2025

**CFO Group RBC Quarterly Team Award** | RBC — Q4 2023